

KRI 7: Threshold Headroom (draft)

Objective

Find ADs where all productive alerts sit well above the threshold, so the threshold can be raised and the false positives below the lowest productive are never generated. A raise fails at exactly one point, the lowest productive value, so the KRI measures that point directly.

The metric

Normalize: $r = \text{alert value} / T$. Then, per AD:

$H = q_alpha(\text{productive } r\text{-values})$
 $R = \# \text{ FP alerts with } r < H$
 $P = \# \text{ productives with } r < H$

how far T can safely be raised

false positives eliminated

detections it costs (0 at alpha = 0)

alpha = tolerated productive loss, set by business per risk tier (high risk: alpha = 0).

Example. T = 10,000; productives at 24,500 / 31,000 / 38,200 / 55,000. At alpha = 0: H = 2.45. Raise T to 24,500, eliminate every FP below it, lose zero observed productives.

Firing condition

FIRE when $H \geq h_min$ AND $R \geq r_min$

(pooled 4-quarter window)

$h_min \sim 1.25$ to 1.5 : minimum headroom worth a tuning cycle. r_min : reclaimed volume whose saved hours exceed the cost of the tuning exercise. Stability: H on each half-window must also clear h_min .

Band	Condition
Green	$H < 1.25$, no action
Amber	$1.25 \leq H < 1.5$, monitor
Red	$H \geq 1.5$ and $R \geq r_min$, tuning candidate

Safeguards

- $n \geq 5$ productives: fire on lower bound of H's confidence interval. $n = 2$ to 4 : indicative, analyst review. $n \leq 1$: separate retirement / BTL review list.
- Recency veto: a productive near T in the latest quarter blocks the candidate.
- Every fired candidate requires BTL validation at the proposed threshold before implementation. The KRI prioritizes; the existing tuning process authorizes.

Output

Quarterly ranked table: AD, H, new T, R, P, hours saved, risk tier, recommendation. One sentence per row: "raise T to H x T, save R alerts (Y hours), at P productives lost."